The time necessary for obtaining the maximum daughter intensity in the non-equilibrium case of the shorter-lived parent is given by

$$t_{max} = (\lambda_2 - \lambda_1)^{-1} \ln(\lambda_2/\lambda_1) \tag{5.83}$$

5.16. RADIOISOTOPE GENERATORS

The growth of radioactive daughters frequently has practical significance. For example, in radiation therapy and diagnostic medicine it is preferable to use short-lived nuclides. In fact, it is preferable to conduct tracer experiments with short-lived nuclides as this eliminates the problem of disposal of residual radioactive waste after completion of the experiment. It is convenient to have a long-lived mother in storage from which a short-lived daughter can be removed as required for use in tracer work. A few examples of uses of such mother-daughter pairs are discussed; others are included in Table 5.1.

Such systems are called *radioisotope generators*. ^{222}Rn is sometimes used for the radiotherapeutic treatment of cancer. This product is isolated by separating it as a gas from the parent substance ^{226}Ra which is normally in the form of solid or a solution of $RaBr_2$.

Table 5.1 Some common radioactive milking pairs. The decay properties include decay energy (MeV), mode of decay and half-life

Mother nuclide	Decay properties	Daughter nuclide	Decay properties	Application
^{44}Ti	EC, γ; 47.3 y	^{44}Sc	1.5 β^+; 1.16 γ; 3.93 h	Teaching
^{68}Ge	EC; 270.8 d	^{68}Ga	1.9 β^+; 1.08 γ; 1.135 h	Medical
^{87}Y	EC; 3.35 d	^{87m}Sr	0.39 γ; 2.80 h	Medical & teaching
^{90}Sr	0.5 β^-; 28.5 y	^{90}Y	2.3 β^-; 2.671 d	Heat source†, Calibration source
^{99}Mo	β^-, γ; 65.9 h	^{99m}Tc	0.14 γ; 6.0 h	Medical
^{113}Sn	EC, γ; 115.1 d	^{113m}In	0.39 γ; 1.658 h	Medical
^{132}Te	β^-, γ; 78.2 h	^{132}I	2.1 β^-, γ; 2.28 h	Medical
^{137}Cs	β^-, γ; 30.0 y	^{137m}Ba	0.66 γ; 2.55 m	Gamma radiography, Radiation sterilization†
^{140}Ba	β^-, γ; 12.75 d	^{140}La	β^-, γ; 1.678 d	Lanthanum tracer
^{144}Ce	β^-, γ; 284.9 d	^{144}Pr	3.0 β^-; 17.28 m	Calibration source
^{210}Pb	β^-, γ; 22.3 y	^{210}Bi	1.2 β^-; 5.01 d	Calibration source
^{226}Ra	α; 1600 y	^{222}Rn	α; 3.825 d	Medical
^{238}U	α; 4.468×10^9 y	^{234}Th	β^-, γ; 24.1 d	Thorium tracer

†Main use of mother substance in large amounts.

^{222}Rn grows into the radium sample with a half-life of 3.8 d. After a 2-week period, following a separation of radon from radium, approximately 90% of the maximum amount of radon has grown back in the radium sample. Consequently, it is useful to separate ^{222}Rn each 2 weeks from the radium samples since further time provides very little additional radioactivity. The ^{222}Rn is an α emitter; the therapeutic value comes from the irradiation of the tissue by the γ-rays of the decay daughters ^{214}Pb and ^{214}Bi which reach radioactive equilibrium extremely rapidly with the ^{222}Rn.

^{99m}Tc is used for diagnostic purposes for liver, spleen, and thyroid scanning (§18.5). The ^{99}Mo parent, obtained by chemical separation from ^{235}U fission product mixture, is absorbed on a column of alumina and the daughter ^{99m}Tc removed by passage of saline solution at intervals governed by the equilibrium. The parent, when it is fixed in a semi-permanent sample as on an adsorbent column, is often known as a cow and the removal of the daughter activity from the radioisotope generator (the "cow") is termed *milking*.

Another commonly used radioisotope generator is ^{132}Te from which ^{132}I may be milked. In this case ^{132}Te is adsorbed as barium tellurite on an alumina column, and the ^{132}I removed by passage of 0.01 M ammonia through the column. The ^{132}I is used both diagnostically and therapeutically for thyroid cancer.

Many of these sources produce radionuclides with half-lives suitable for teaching purposes, e.g.

$$^{137m}\text{Ba}(2.6\text{ min}),\ ^{144}\text{Pr}(17.3\text{ min}),\ ^{44}\text{Sc}(4.0\text{ h}),\ ^{99m}\text{Tc}(6.0\text{ h}) \text{ and } ^{90}\text{Y}(64\text{ h}). \quad (5.84)$$

5.17. DECAY ENERGY AND HALF-LIFE

It was observed early in both α- and β-decay that the longer the half-life the lower the decay energy. Although there are many exceptions to this observation H. Geiger and J. M. Nuttall formulated the law

$$\log \lambda_\alpha = a + b \log \hat{R}_{\text{air}} \quad (5.85)$$

for the natural α-active nuclides. Here a and b are constants, and $\hat{R}_{\text{air}}$ is the range of the α-particles in air which is directly proportional to the α-particle energy E_α. A similar relation was deduced by E. Fermi for the β-decay:

$$\log \lambda_\beta = a' + b' \log E \quad (5.86)$$

where a' is a constant related to the type of β decay and $b' \approx 5$.

Although these rules have been superseded by modern theory and the enormous amount of nuclear data now available, they may nevertheless be useful as rough guides in estimates of half-lives and decay energies. In §6.7 more valid but also more complicated relationships are discussed.